```
Starting ML pipeline

==============================
Running: src/01_get_pollutants_data.py
==============================
API_KEY loaded: True
TIME WINDOW: 2023-03-02 20:00:00+00:00 → 2026-03-01 18:00:00+00:00
Locations found: 7
Sensors selected: 33
==============================
API_KEY loaded: True
TIME WINDOW: 2023-03-02 20:00:00+00:00 → 2026-03-01 18:00:00+00:00
Locations found: 7
Sensors selected: 33
Downloading sensors:   3%|[redacted]| 1/33 [00:16<08:47, 16.48s/it][RateLimit] waiting 17s...
Downloading sensors:  18%|[redacted]| 6/33 [03:54<18:02, 40.08s/it][RateLimit] waiting 4s...
Downloading sensors:  21%|[redacted]| 7/33 [04:16<14:47, 34.14s/it][RateLimit] waiting 9s...
Downloading sensors:  36%|[redacted]| 12/33 [07:31<11:54, 34.01s/it][RateLimit] waiting 26s...
```

Downloading sensors: 42%|[redacted] | 14/33 [08:31<09:47, 30.91s/it][RateLimit] waiting 35s...

Downloading sensors: 45%|[redacted] | 15/33 [09:23<11:10, 37.23s/it][RateLimit] waiting 36s...

Downloading sensors: 52%|[redacted] | 17/33 [10:31<09:03, 33.98s/it][RateLimit] waiting 35s...

Downloading sensors: 55%|[redacted] | 18/33 [11:23<09:49, 39.28s/it][RateLimit] waiting 33s...

Downloading sensors: 61%|[redacted] | 20/33 [12:31<07:37, 35.17s/it][RateLimit] waiting 33s...

Downloading sensors: 67%|[redacted] | 22/33 [13:40<06:05, 33.22s/it][RateLimit] waiting 33s...

Downloading sensors: 70%|[redacted] | 23/33 [14:30<06:22, 38.28s/it][RateLimit] waiting 23s...

Downloading sensors: 79%|[redacted] | 26/33 [16:42<04:44, 40.66s/it][RateLimit] waiting 4s...

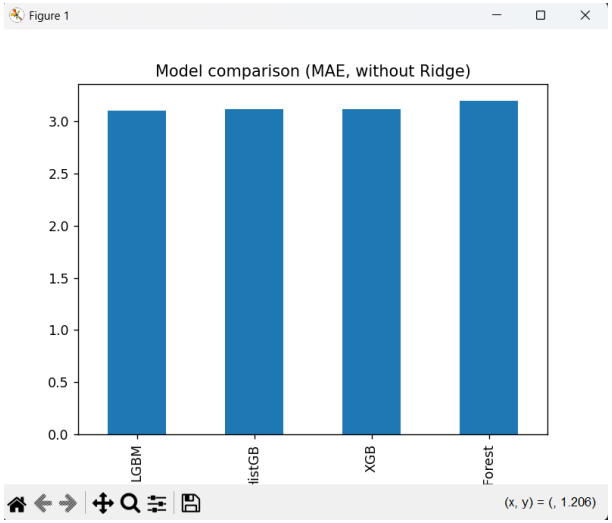
Downloading sensors: 85%|[redacted] | 28/33 [17:57<03:13, 38.64s/it][RateLimit] waiting 5s...

Downloading sensors: 88%|[redacted] | 29/33 [18:41<02:41, 40.32s/it][RateLimit] waiting 14s...

```
Downloading sensors:
94%|[redacted]| 31/33 [19:37<01:07, 33.77s/it][RateLimit] waiting 20s...

Downloading sensors:
100%|[redacted]| 33/33 [20:48<00:00, 37.84s/it]

Saved → ./data/raw/pollutants.csv


==============================

Running: src/02_get_weather_data.py

==============================

Downloading weather data...

TIME WINDOW: 2023-03-02 20:00:00+00:00 → 2026-03-01 18:00:00+00:00

Saved weather dataset.


==============================

Running: src/03_preprocess.py

==============================

Loading raw datasets...

Pollutants shape: (76700, 10)

Weather shape: (26279, 6)
```

```
Building base dataset...

Building features...
Train dataset with features: (75647, 39)
Test dataset with features: (25216, 40)

Datasets saved.
Train dataset shape: (75647, 38)
Time range: 2023-03-19 00:00:00+00:00 → 2025-06-12 01:00:00+00:00
Test dataset shape: (25216, 39)
Time range: 2025-06-12 02:00:00+00:00 → 2026-03-01 17:00:00+00:00

Saved feature list.

==============================
Running: src/04_train.py
==============================
Train: (34836, 34)
Test : (22125, 34)
```

```
Train period: 2023-03-20 00:00:00+00:00 → 2025-06-12 00:00:00+00:00
Test period : 2025-06-13 02:00:00+00:00 → 2026-03-01 16:00:00+00:00

Training models...
[LightGBM] [Info] Auto-choosing col-wise multi-threading, the overhead of testing was 0.002018 seconds.
You can set `force_col_wise=true` to remove the overhead.
[LightGBM] [Info] Total Bins 5216
[LightGBM] [Info] Number of data points in the train set: 34836, number of used features: 34
[LightGBM] [Info] Start training from score 2.576769
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
[LightGBM] [Warning] No further splits with positive gain, best gain: -inf
Top models: ['LGBM', 'HistGB']
        model        MAE           R2
2        LGBM   3.103176  8.374272e-01
1      HistGB   3.114594  8.360043e-01
3         XGB   3.119178  8.390235e-01
0  RandomForest   3.197240  8.283223e-01
4       Ridge  930.534002 -6.392484e+07
```



Best model = LGBM saved.

Top-2 ensemble saved.

Best model: LGBM

=== OVERFITTING CHECK ===

Train MAE: 2.024 µg/m³

Test MAE: 3.103 µg/m³

Train R2 : 0.935

Test R2 : 0.837

```
==============================
Running: src/05_evaluate.py
==============================
Model loaded:  ./models/model.pkl
Dataset loaded: (75647, 38)

Validation results
1-step MAE: 3.103 µg/m³
RMSE : 6.65
MAPE : 0.208
SMAPE: 0.195
MASE : 0.685
R2   : 0.837
```

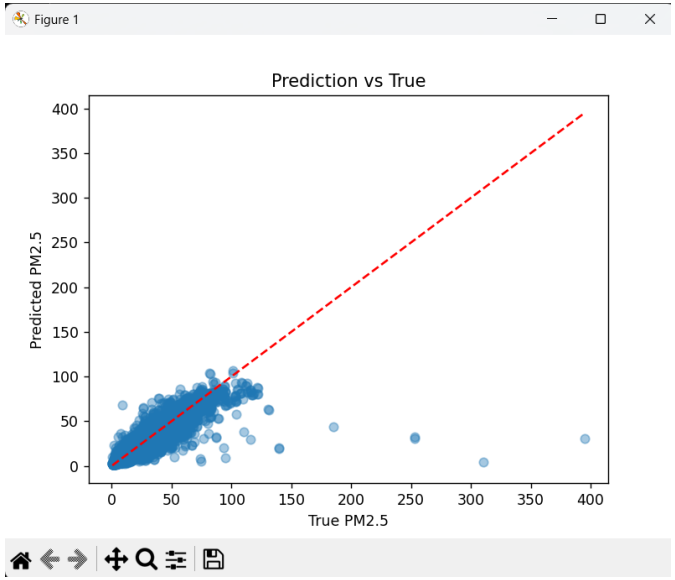


Figure 1
Prediction vs True
Predicted PM2.5
True PM2.5
0
50
100
150
200
250
300
350
400

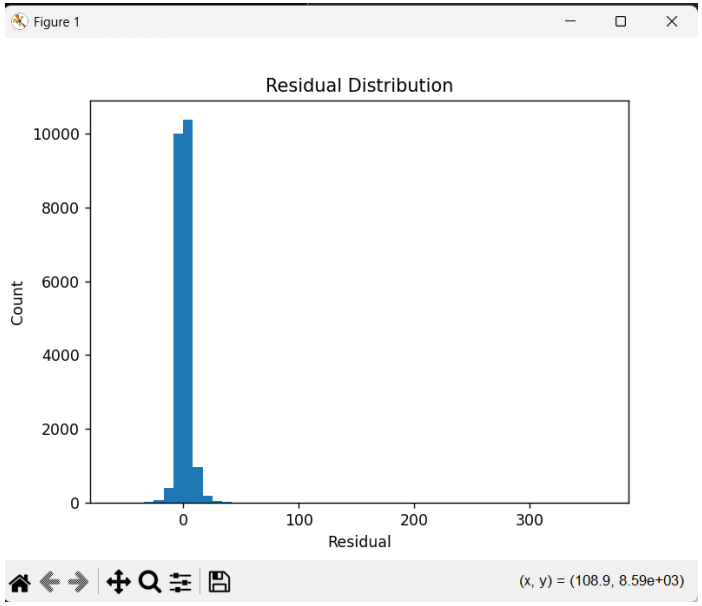


Figure 1
Residual Distribution
Count
Residual
0
2000
4000
6000
8000
10000
100
200
300
(x, y) = (108.9, 8.59e+03)

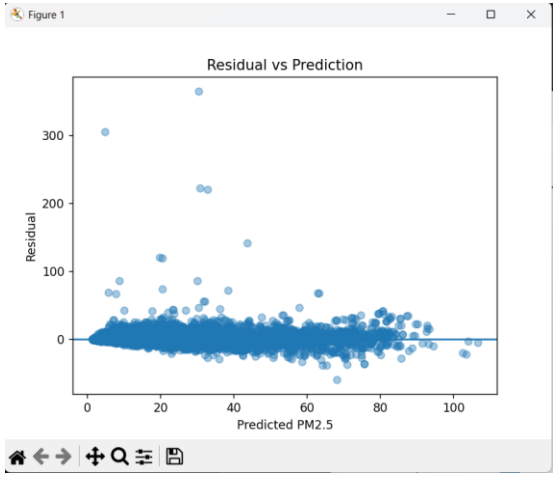


Figure 1
Residual vs Prediction
Residual
300
200
100
0
0
20
40
60
80
100
Predicted PM2.5

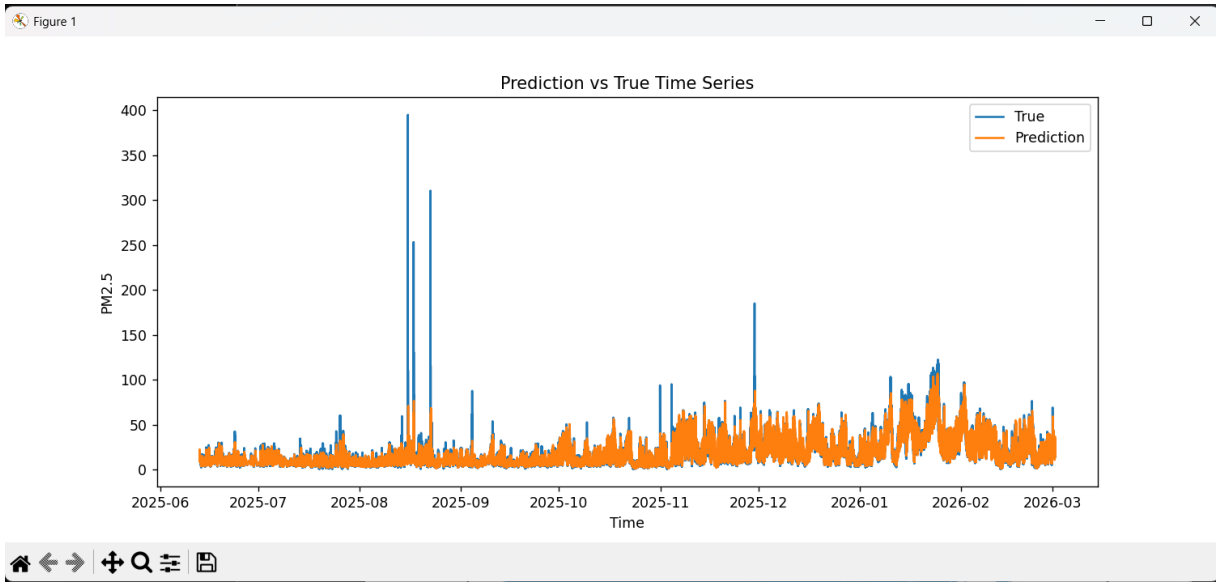


Figure 1
Prediction vs True Time Series
True
Prediction
PM2.5
400
350
300
250
200
150
100
50
0
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02
2026-03
Time

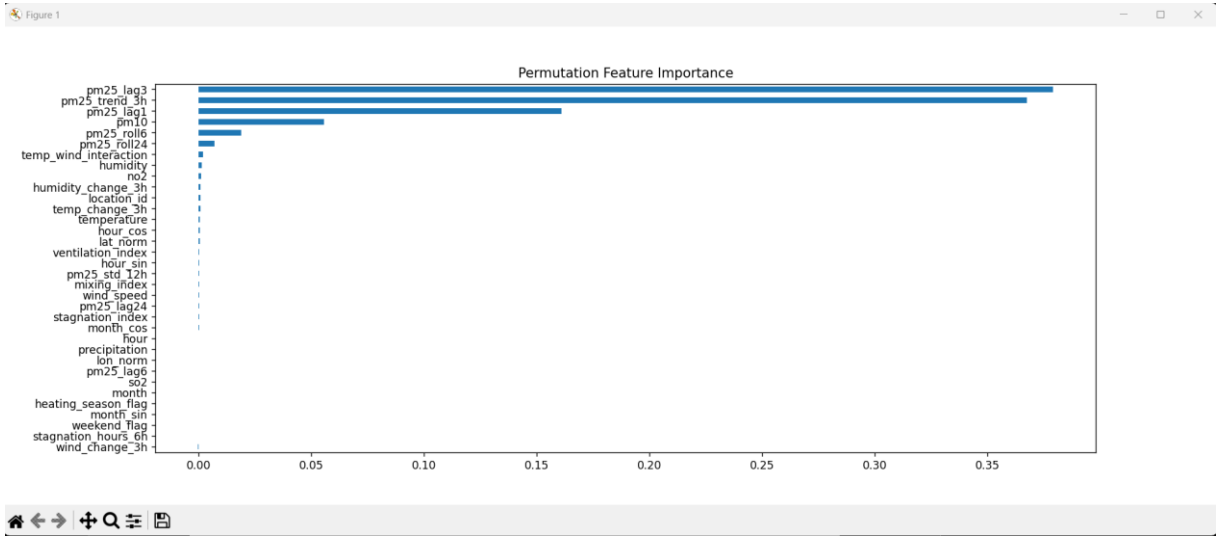


Figure 1
Permutation Feature Importance
pm25_lag3
pm25_trend_3h
pm25_lag1
pm10
pm25_roll6
pm25_roll24
temp_wind_interaction
humidity
no2
humidity_change_3h
location_id
temp_change_3h
temperature
hour_cos
lat_norm
ventilation_index
hour_sin
pm25_std_12h
mixing_index
wind_speed
pm25_lag24
stagnation_index
month_cos
hour
precipitation
lon_norm
pm25_lag6
so2
month
heating_season_flag
month_sin
weekend_flag
stagnation_hours_6h
wind_change_3h
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35

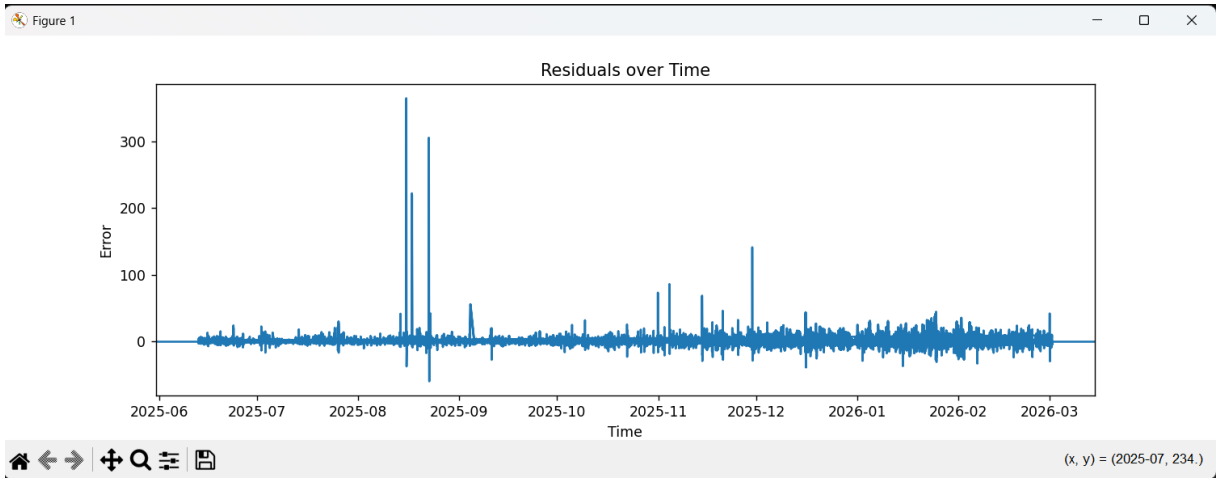
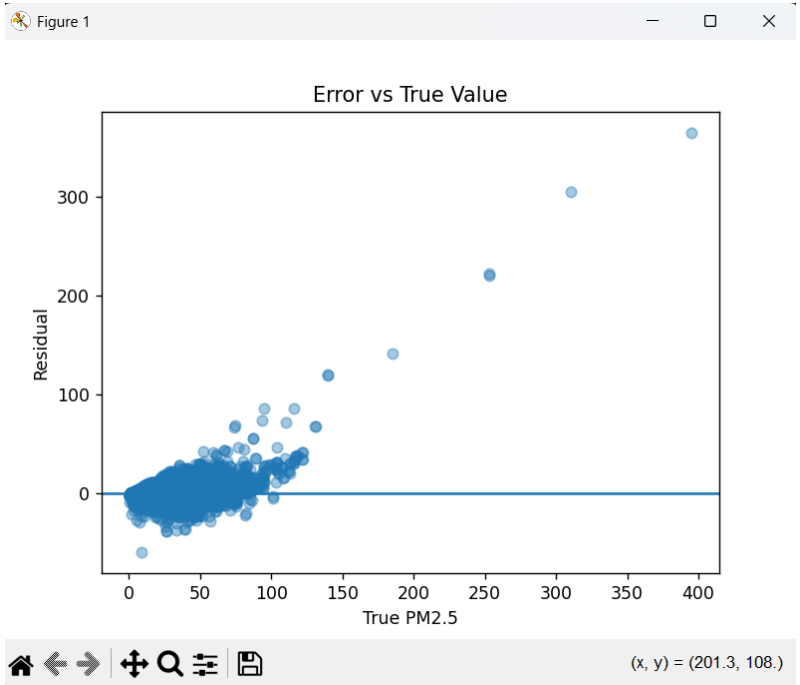


Figure 1
Residuals over Time
Error
300
200
100
0
2025-06
2025-07
2025-08
2025-09
2025-10
2025-11
2025-12
2026-01
2026-02
2026-03
Time
(x, y) = (2025-07, 234.)



Running SHAP analysis...

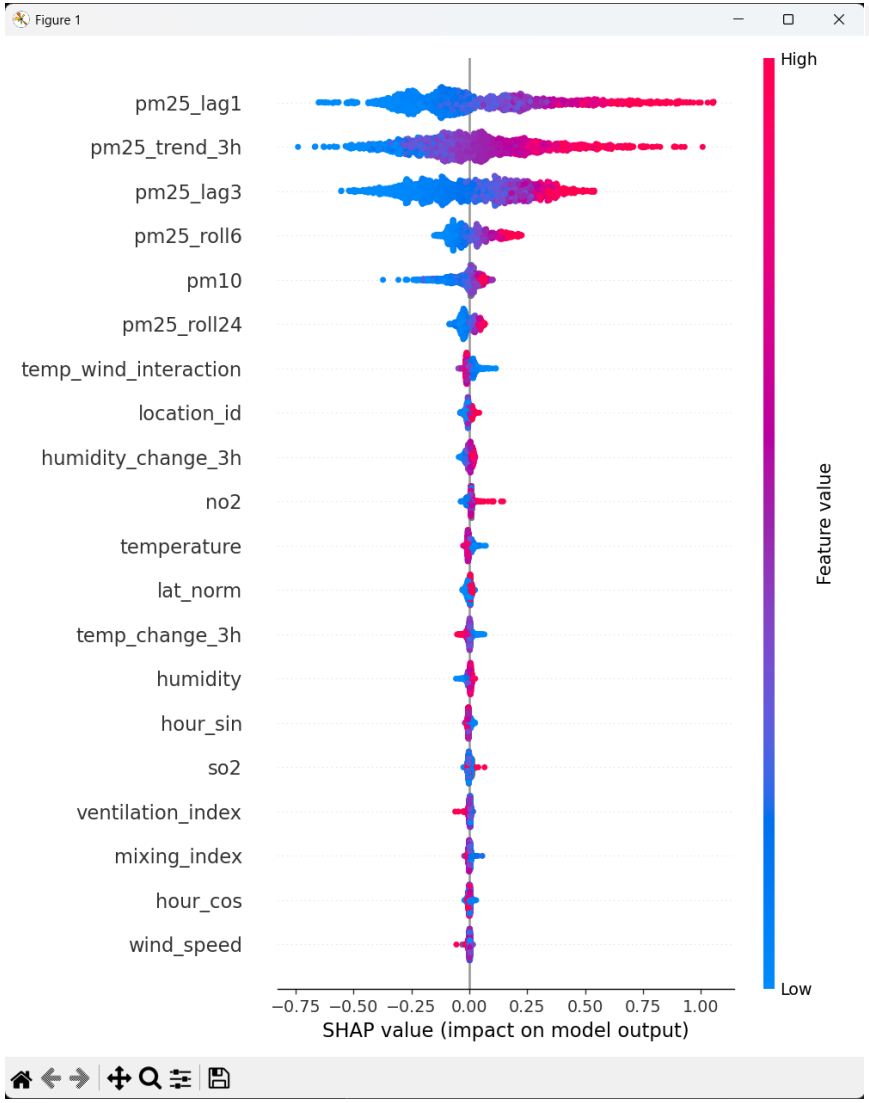


Figure 1
pm25_lag1
pm25_trend_3h
pm25_lag3
pm25_roll6
pm10
pm25_roll24
temp_wind_interaction
location_id
humidity_change_3h
no2
temperature
lat_norm
temp_change_3h
humidity
hour_sin
so2
ventilation_index
mixing_index
hour_cos
wind_speed
−0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00
SHAP value (impact on model output)
High
Feature value
Low

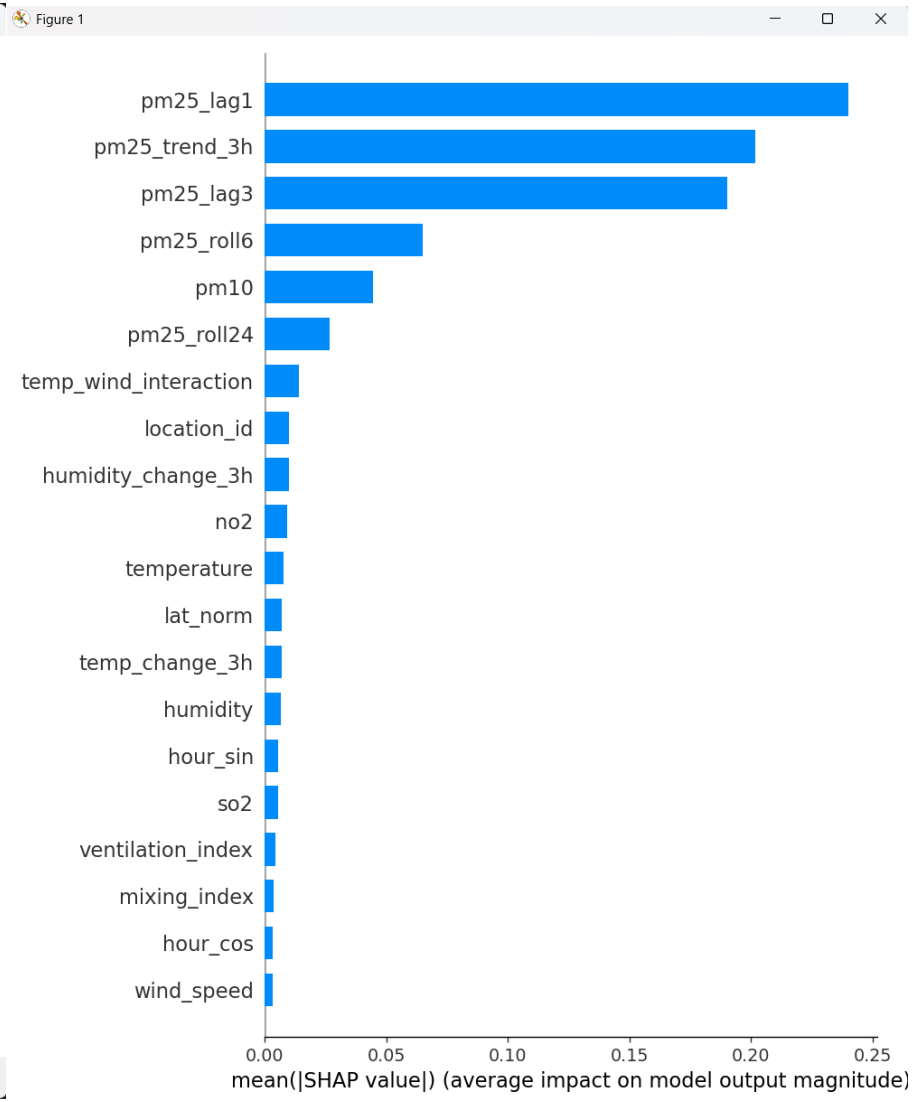


Figure 1
pm25_lag1
pm25_trend_3h
pm25_lag3
pm25_roll6
pm10
pm25_roll24
temp_wind_interaction
location_id
humidity_change_3h
no2
temperature
lat_norm
temp_change_3h
humidity
hour_sin
so2
ventilation_index
mixing_index
hour_cos
wind_speed
0.00 0.05 0.10 0.15 0.20 0.25
mean(|SHAP value|) (average impact on model output magnitude)

```
Top features (SHAP):
pm25_lag1              0.239979
pm25_trend_3h          0.201614
pm25_lag3              0.190078
pm25_roll6             0.064974
pm10                   0.044687
pm25_roll24            0.026881
temp_wind_interaction  0.014270
location_id            0.010028
humidity_change_3h     0.009936
no2                    0.009474
dtype: float64
Metrics saved to ./models/evaluation.json

==============================
Running: src/06_forecast.py
==============================
[2026-03-31 10:37:17] [INIT] Modell és feature lista betöltése
[2026-03-31 10:37:18] [INIT] Features count: 34
[2026-03-31 10:37:18] [DATA] Pollution letöltés indul
```

Downloading pollutants: 2026-03-29 08:37:18.409047+00:00 → 2026-03-31 08:37:18.409047+00:00

Location: Gyor Szent Istvan

Fetching co | sensor 9308

Fetching no2 | sensor 9307

Fetching pm10 | sensor 9306

Fetching pm25 | sensor 36004

Fetching so2 | sensor 9345

[2026-03-31 10:37:20] [DATA] Pollution shape: (47, 9)

[2026-03-31 10:37:20] [DATA] Pollution head:

parameter	datetime	co	no2	pm10	pm25	so2	location	latitude	longitude
44	2026-03-31 04:00:00+00:00	499.524	6.485301	9.196459	8.479122	-1.0	Gyor Szent Istvan	47.685278	17.639444
45	2026-03-31 05:00:00+00:00	530.061	11.151095	12.550992	11.345555	-1.0	Gyor Szent Istvan	47.685278	17.639444
46	2026-03-31 06:00:00+00:00	548.388	13.990507	12.464540	11.571052	-1.0	Gyor Szent Istvan	47.685278	17.639444

[2026-03-31 10:37:20] [DATA] Weather history letöltés

Downloading historical weather...

[2026-03-31 10:37:21] [DATA] Weather hist shape: (51, 5)

[2026-03-31 10:37:21] [DEBUG] [DATA] Weather forecast letöltés

Downloading weather forecast...

[2026-03-31 10:37:21] [DATA] Weather forecast range: 2026-03-31 00:00:00+00:00 → 2026-04-01 23:00:00+00:00

[2026-03-31 10:37:21] [PIPELINE] Base dataset építés

[2026-03-31 10:37:21] [PIPELINE] History shape: (47, 12)

[2026-03-31 10:37:21] [PIPELINE] Last rows:

	location	datetime	no2	pm10	pm25	so2	temperature	humidity	wind_speed	precipitation	latitude	longitude
44	Gyor Szent Istvan	2026-03-31 04:00:00+00:00	6.485301	9.196459	8.479122	NaN	4.6	86.0	20.1	0.0	47.685278	17.639444
45	Gyor Szent Istvan	2026-03-31 05:00:00+00:00	11.151095	12.550992	11.345555	NaN	4.7	86.0	21.2	0.0	47.685278	17.639444
46	Gyor Szent Istvan	2026-03-31 06:00:00+00:00	13.990507	12.464540	11.571052	NaN	5.2	84.0	23.2	0.1	47.685278	17.639444

[2026-03-31 10:37:21] [FORECAST] Start time: 2026-03-31 06:00:00+00:00

[2026-03-31 10:37:21] [DEBUG]

==========

[2026-03-31 10:37:21] [DEBUG] [STEP 1]

[2026-03-31 10:37:21] [DEBUG] Future time: 2026-03-31 07:00:00+00:00

[2026-03-31 10:37:21] [DEBUG] Last known pm25: 11.5710516

[2026-03-31 10:37:21] [DEBUG] Weather values: T=5.7, H=81.0, W=20.6, P=0.0

[2026-03-31 10:37:21] [DEBUG] History length: 48

[2026-03-31 10:37:21] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 11.5710516, 'pm25_lag3': 8.479122, 'pm25_lag6': 4.121652, 'pm25_lag24': 9.746299, 'pm25_roll6': 7.686931650000001, 'pm25_roll24': 7.844053983333333, 'pm25_trend_3h': 3.0919296000000003, 'pm25_std_12h': 2.44510940916836, 'hour': 7.0, 'hour_sin': 0.9659258262890683, 'hour_cos': -0.25881904510252063, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 5.7, 'humidity': 81.0, 'wind_speed': 20.6, 'precipitation': 0.0, 'temp_change_3h': 1.1000000000000005, 'humidity_change_3h': -5.0, 'wind_change_3h': 0.5, 'stagnation_index': 3.838862559241706, 'temp_wind_interaction': 117.42000000000002, 'mixing_index': 59.63524752475247, 'ventilation_index': 5744.3099999999995, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:21] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 10.056236371247723

[2026-03-31 10:37:23] [DEBUG]

==========

[2026-03-31 10:37:23] [DEBUG] [STEP 2]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 08:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 10.056236371247723

[2026-03-31 10:37:23] [DEBUG] Weather values: T=6.3, H=77.0, W=21.7, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 49

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 10.056236371247723, 'pm25_lag3': 11.3455553, 'pm25_lag6': 4.456957, 'pm25_lag24': 10.7824221, 'pm25_roll6': 8.676029045207953, 'pm25_roll24': 7.856968040468654, 'pm25_trend_3h': -1.2893189287522766, 'pm25_std_12h': 2.5333850041817536, 'hour': 8.0, 'hour_sin': 0.8660254037844387, 'hour_cos': -0.4999999999999998, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 6.3, 'humidity': 77.0, 'wind_speed': 21.7, 'precipitation': 0.0, 'temp_change_3h': 1.5999999999999996, 'humidity_change_3h': -9.0, 'wind_change_3h': 0.5, 'stagnation_index': 3.4684684684684686, 'temp_wind_interaction': 136.70999999999998, 'mixing_index': 65.3970618556701, 'ventilation_index': 6064.065, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 9.34076284719846

[2026-03-31 10:37:23] [DEBUG]

==========

[2026-03-31 10:37:23] [DEBUG] [STEP 3]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 09:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 9.34076284719846

[2026-03-31 10:37:23] [DEBUG] Weather values: T=6.9, H=71.0, W=22.8, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 50

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 9.34076284719846, 'pm25_lag3': 11.5710516, 'pm25_lag6': 6.147252, 'pm25_lag24': 9.867472, 'pm25_roll6': 9.489996686407698, 'pm25_roll24': 7.796898904935257, 'pm25_trend_3h': -2.23028875280154, 'pm25_std_12h': 2.513733561732599, 'hour': 9.0, 'hour_sin': 0.7071067811865476, 'hour_cos': -0.7071067811865475, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 6.9, 'humidity': 71.0, 'wind_speed': 22.8, 'precipitation': 0.0, 'temp_change_3h': 1.7000000000000002, 'humidity_change_3h': -13.0, 'wind_change_3h': -0.3999999999999986, 'stagnation_index': 3.0472103004291844, 'temp_wind_interaction': 157.32000000000002, 'mixing_index': 73.24384615384614, 'ventilation_index': 6385.139999999999, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 8.310808452166766

[2026-03-31 10:37:23] [DEBUG]

==========

[2026-03-31 10:37:23] [DEBUG] [STEP 4]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 10:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 8.310808452166766

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.1, H=70.0, W=23.7, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 51

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 8.310808452166766, 'pm25_lag3': 10.056236371247723, 'pm25_lag6': 8.479122, 'pm25_lag24': 9.5544, 'pm25_roll6': 9.850589428435491, 'pm25_roll24': 7.73203792377554, 'pm25_trend_3h': -1.7454279190809565, 'pm25_std_12h': 2.4797559597528562, 'hour': 10.0, 'hour_sin': 0.49999999999999994, 'hour_cos': -0.8660254037844387, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.1, 'humidity': 70.0, 'wind_speed': 23.7, 'precipitation': 0.0, 'temp_change_3h': 1.3999999999999995, 'humidity_change_3h': -11.0, 'wind_change_3h': 3.099999999999998, 'stagnation_index': 2.8925619834710745, 'temp_wind_interaction': 168.26999999999998, 'mixing_index': 76.91305555555556, 'ventilation_index': 6641.925, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.805695017804963

[2026-03-31 10:37:23] [DEBUG]

==========

[2026-03-31 10:37:23] [DEBUG] [STEP 5]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 11:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.805695017804963

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.6, H=69.0, W=23.6, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 52

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.805695017804963, 'pm25_lag3': 9.34076284719846, 'pm25_lag6': 11.3455553, 'pm25_lag24': 9.154694, 'pm25_roll6': 9.73835159806965, 'pm25_roll24': 7.659175216184079, 'pm25_trend_3h': -1.5350678293934976, 'pm25_std_12h': 2.477875052494111, 'hour': 11.0, 'hour_sin': 0.258819045102521, 'hour_cos': -0.9659258262890682, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.6, 'humidity': 69.0, 'wind_speed': 23.6, 'precipitation': 0.0, 'temp_change_3h': 1.2999999999999998, 'humidity_change_3h': -8.0, 'wind_change_3h': 1.9000000000000021, 'stagnation_index': 2.863070539419087, 'temp_wind_interaction': 179.36, 'mixing_index': 77.60056179775282, 'ventilation_index': 6625.700000000001, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.293138618102516

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 6]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 12:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.293138618102516

[2026-03-31 10:37:23] [DEBUG] Weather values: T=8.1, H=67.0, W=23.7, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 53

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.293138618102516, 'pm25_lag3': 8.310808452166766, 'pm25_lag6': 11.5710516, 'pm25_lag24': 10.2676468, 'pm25_roll6': 9.062948817753405, 'pm25_roll24': 7.581610408605019, 'pm25_trend_3h': -1.0176698340642503, 'pm25_std_12h': 2.473524166590683, 'hour': 12.0, 'hour_sin': 1.2246467991473532e-16, 'hour_cos': -1.0, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 8.1, 'humidity': 67.0, 'wind_speed': 23.7, 'precipitation': 0.0, 'temp_change_3h': 1.1999999999999993, 'humidity_change_3h': -4.0, 'wind_change_3h': 0.8999999999999986, 'stagnation_index': 2.768595041322314, 'temp_wind_interaction': 191.97, 'mixing_index': 79.84913793103448, 'ventilation_index': 6665.625, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.35860280400437

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 7]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 13:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.35860280400437

[2026-03-31 10:37:23] [DEBUG] Weather values: T=8.1, H=67.0, W=23.0, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 54

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.35860280400437, 'pm25_lag3': 7.805695017804963, 'pm25_lag6': 10.056236371247723, 'pm25_lag24': 13.1101418, 'pm25_roll6': 8.3608740184208, 'pm25_roll24': 7.460400242105199, 'pm25_trend_3h': -0.44709221380059283, 'pm25_std_12h': 2.379198037611623, 'hour': 13.0, 'hour_sin': -0.2588190451025208, 'hour_cos': -0.9659258262890683, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 8.1, 'humidity': 67.0, 'wind_speed': 23.0, 'precipitation': 0.0, 'temp_change_3h': 1.0, 'humidity_change_3h': -3.0, 'wind_change_3h': -0.6999999999999993, 'stagnation_index': 2.851063829787234, 'temp_wind_interaction': 186.29999999999998, 'mixing_index': 77.58620689655173, 'ventilation_index': 6468.75, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.316251943258452

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 8]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 14:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.316251943258452

[2026-03-31 10:37:23] [DEBUG] Weather values: T=8.1, H=64.0, W=21.3, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 55

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.316251943258452, 'pm25_lag3': 7.293138618102516, 'pm25_lag6': 9.34076284719846, 'pm25_lag24': 9.128088, 'pm25_roll6': 7.904209947089254, 'pm25_roll24': 7.21898816474097, 'pm25_trend_3h': 0.023113325155936337, 'pm25_std_12h': 2.0602111277768698, 'hour': 14.0, 'hour_sin': -0.4999999999999997, 'hour_cos': -0.8660254037844388, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 8.1, 'humidity': 64.0, 'wind_speed': 21.3, 'precipitation': 0.0, 'temp_change_3h': 0.5, 'humidity_change_3h': -5.0, 'wind_change_3h': -2.3000000000000007, 'stagnation_index': 2.935779816513761, 'temp_wind_interaction': 172.53, 'mixing_index': 74.66517857142857, 'ventilation_index': 5990.625, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.420875943755133

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 9]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 15:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.420875943755133

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.9, H=68.0, W=20.8, P=0.1

[2026-03-31 10:37:23] [DEBUG] History length: 56

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.420875943755133, 'pm25_lag3': 7.35860280400437, 'pm25_lag6': 8.310808452166766, 'pm25_lag24': 5.421771, 'pm25_roll6': 7.584228796515366, 'pm25_roll24': 7.147854329064098, 'pm25_trend_3h': 0.06227313975076321, 'pm25_std_12h': 1.7061266731695597, 'hour': 15.0, 'hour_sin': -0.7071067811865471, 'hour_cos': -0.7071067811865479, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.9, 'humidity': 68.0, 'wind_speed': 20.8, 'precipitation': 0.1, 'temp_change_3h': -0.1999999999999993, 'humidity_change_3h': 1.0, 'wind_change_3h': -2.8999999999999986, 'stagnation_index': 3.192488262910798, 'temp_wind_interaction': 164.32000000000002, 'mixing_index': 69.62374999999999, 'ventilation_index': 5845.839999999999, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.7000822481424525

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 10]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 16:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.7000822481424525

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.5, H=66.0, W=21.0, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 57

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.7000822481424525, 'pm25_lag3': 7.316251943258452, 'pm25_lag6': 7.805695017804963, 'pm25_lag24': 7.150046, 'pm25_roll6': 7.482441095844647, 'pm25_roll24': 7.242783964403369, 'pm25_trend_3h': 0.3838303048840004, 'pm25_std_12h': 1.5611125406050628, 'hour': 16.0, 'hour_sin': -0.8660254037844385, 'hour_cos': -0.5000000000000004, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.5, 'humidity': 66.0, 'wind_speed': 21.0, 'precipitation': 0.0, 'temp_change_3h': -0.5999999999999996, 'humidity_change_3h': -1.0, 'wind_change_3h': -2.0, 'stagnation_index': 3.0697674418604652, 'temp_wind_interaction': 157.5, 'mixing_index': 71.79418604651163, 'ventilation_index': 5893.65, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 7.944287719222279

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 11]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 17:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 7.944287719222279

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.4, H=69.0, W=18.3, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 58

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 7.944287719222279, 'pm25_lag3': 7.420875943755133, 'pm25_lag6': 7.293138618102516, 'pm25_lag24': 5.845617, 'pm25_roll6': 7.505539879414201, 'pm25_roll24': 7.2758773693709635, 'pm25_trend_3h': 0.5234117754671459, 'pm25_std_12h': 1.574526049579268, 'hour': 17.0, 'hour_sin': -0.9659258262890683, 'hour_cos': -0.25881904510252063, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.4, 'humidity': 69.0, 'wind_speed': 18.3, 'precipitation': 0.0, 'temp_change_3h': -0.6999999999999993, 'humidity_change_3h': 5.0, 'wind_change_3h': -3.0, 'stagnation_index': 3.6702127659574466, 'temp_wind_interaction': 135.42000000000002, 'mixing_index': 60.83837078651684, 'ventilation_index': 5134.065, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

[2026-03-31 10:37:23] [DEBUG] Pred value: 8.304105797709315

[2026-03-31 10:37:23] [DEBUG]

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[2026-03-31 10:37:23] [DEBUG] [STEP 12]

[2026-03-31 10:37:23] [DEBUG] Future time: 2026-03-31 18:00:00+00:00

[2026-03-31 10:37:23] [DEBUG] Last known pm25: 8.304105797709315

[2026-03-31 10:37:23] [DEBUG] Weather values: T=7.0, H=70.0, W=16.6, P=0.0

[2026-03-31 10:37:23] [DEBUG] History length: 59

[2026-03-31 10:37:23] [DEBUG] Feature vector:

{'pm10': 12.46454, 'no2': 13.9905071, 'so2': nan, 'pm25_lag1': 8.304105797709315, 'pm25_lag3': 7.7000822481424525, 'pm25_lag6': 7.35860280400437, 'pm25_lag24': 4.912048, 'pm25_roll6': 7.674034409348667, 'pm25_roll24': 7.378314402608851, 'pm25_trend_3h': 0.6040235495668629, 'pm25_std_12h': 1.3205563686934285, 'hour': 18.0, 'hour_sin': -1.0, 'hour_cos': -1.8369701987210297e-16, 'month': 3.0, 'month_sin': 1.0, 'month_cos': 6.123233995736766e-17, 'weekend_flag': 0.0, 'heating_season_flag': 1.0, 'temperature': 7.0, 'humidity': 70.0, 'wind_speed': 16.6, 'precipitation': 0.0, 'temp_change_3h': -0.9000000000000004, 'humidity_change_3h': 2.0, 'wind_change_3h': -4.199999999999999, 'stagnation_index': 4.093567251461988, 'temp_wind_interaction': 116.20000000000002, 'mixing_index': 54.784888888888894, 'ventilation_index': 4650.49, 'stagnation_hours_6h': 0.0, 'location_id': 3.0, 'lat_norm': -0.010734629783662797, 'lon_norm': -0.017517650295470588}

[2026-03-31 10:37:23] [WARN] NaN a feature vectorban!

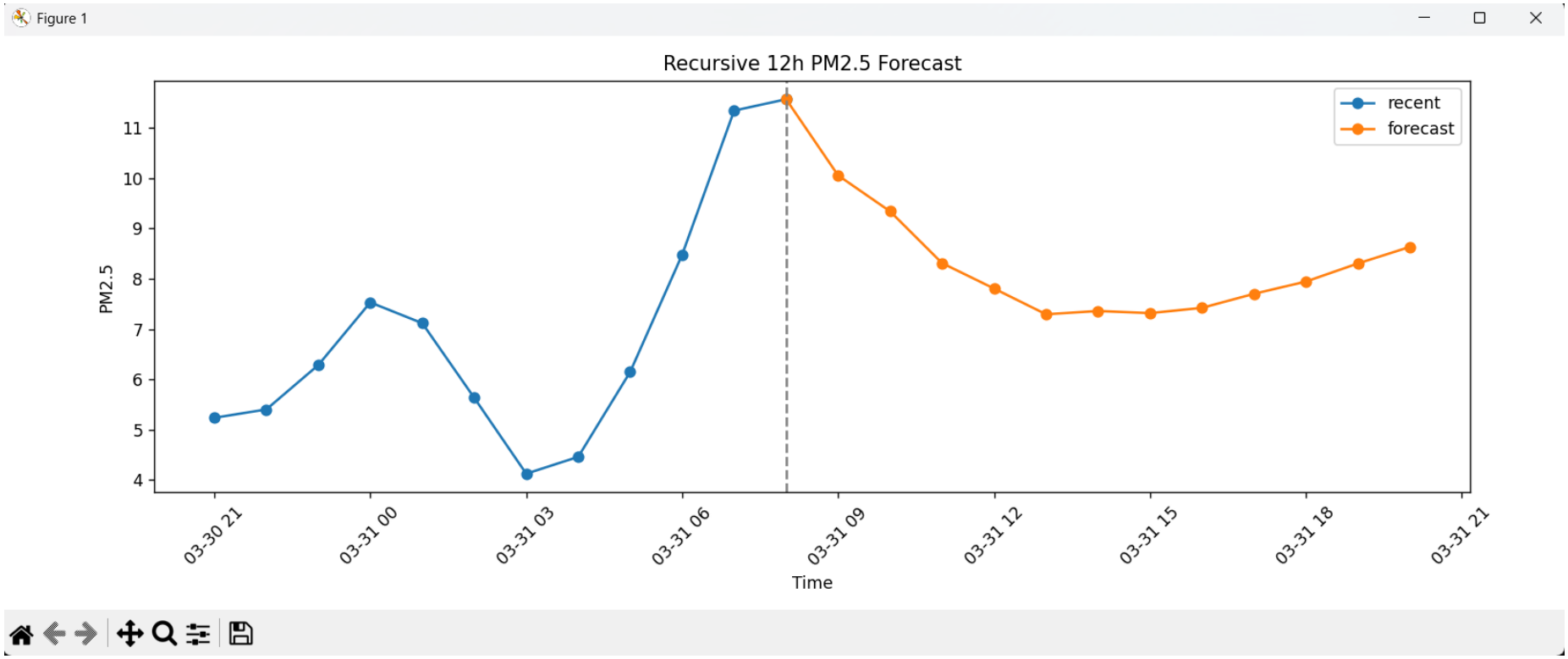
[2026-03-31 10:37:23] [DEBUG] Pred value: 8.63323587990958

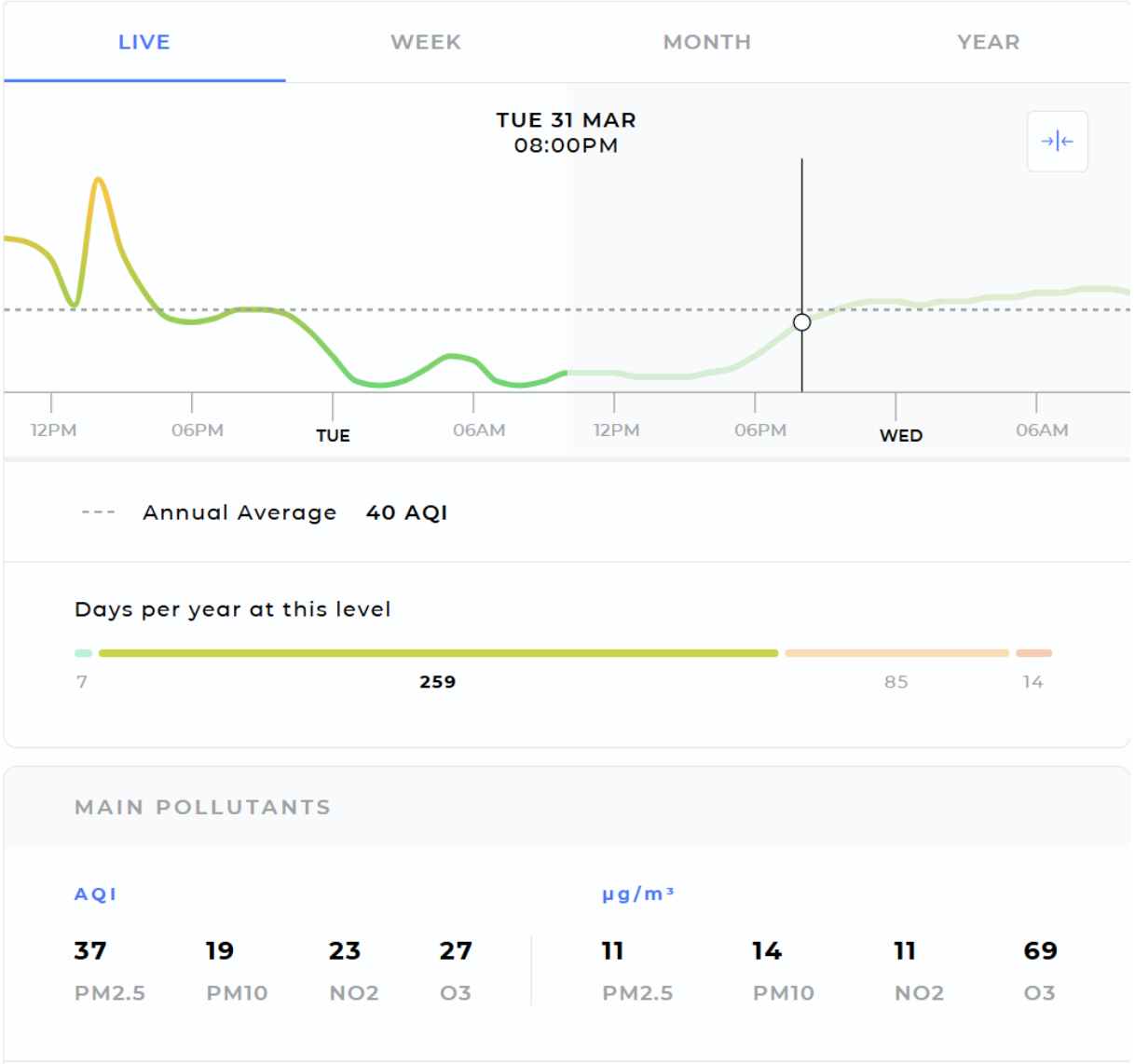
[2026-03-31 10:37:23] [OUTPUT] Forecast:

	datetime	location	pm25_pred	datetime_local
0	2026-03-31 07:00:00+00:00	Gyor Szent Istvan	10.056236	2026-03-31 09:00:00+02:00
1	2026-03-31 08:00:00+00:00	Gyor Szent Istvan	9.340763	2026-03-31 10:00:00+02:00
2	2026-03-31 09:00:00+00:00	Gyor Szent Istvan	8.310808	2026-03-31 11:00:00+02:00
3	2026-03-31 10:00:00+00:00	Gyor Szent Istvan	7.805695	2026-03-31 12:00:00+02:00
4	2026-03-31 11:00:00+00:00	Gyor Szent Istvan	7.293139	2026-03-31 13:00:00+02:00
5	2026-03-31 12:00:00+00:00	Gyor Szent Istvan	7.358603	2026-03-31 14:00:00+02:00
6	2026-03-31 13:00:00+00:00	Gyor Szent Istvan	7.316252	2026-03-31 15:00:00+02:00
7	2026-03-31 14:00:00+00:00	Gyor Szent Istvan	7.420876	2026-03-31 16:00:00+02:00
8	2026-03-31 15:00:00+00:00	Gyor Szent Istvan	7.700082	2026-03-31 17:00:00+02:00
9	2026-03-31 16:00:00+00:00	Gyor Szent Istvan	7.944288	2026-03-31 18:00:00+02:00
10	2026-03-31 17:00:00+00:00	Gyor Szent Istvan	8.304106	2026-03-31 19:00:00+02:00

11 2026-03-31 18:00:00+00:00 Gyor Szent Istvan 8.633236 2026-03-31 20:00:00+02:00

[2026-03-31 10:37:23] [DEBUG] [OUTPUT] CSV mentve





https://air.plumelabs.com/air-quality-in-Gy%C5%91r-cNXX